Passive Inference and Active Clarification:
A Controlled Benchmark for Ask-or-Act Under Instruction Ambiguity

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Abstract. We contribute a controlled benchmark for the ask-or-act decision in cooperative assistance. A reproducible gridworld pairs a Boltzmann principal (whose actions reveal goal information via inverse planning) with an explicit question bank at cost c_q, and varies ambiguity K and principal rationality β systematically. Across 19,800 episodes the two channels exhibit a *strategic-substitutes* pattern: a cost-sensitive Bayes-risk rule gains +7.9–+8.4 pp over not asking at $K \geq 3$; an information-gain policy achieves higher peak success at roughly twice the question cost. To our knowledge, no prior benchmark provides both channels in one controlled environment. Results characterise post-parsing instruction ambiguity in this small gridworld and Boltzmann principal.

1. Introduction

A central challenge in cooperative assistance is deciding when a helper should ask for clarification rather than act. Acting immediately risks committing to the wrong goal, whereas asking too often wastes time and may frustrate the principal. The trade-off depends on the informativeness of available questions, the cost of a wrong action, and what the principal's behaviour already reveals.

We study this question by constructing an attribute-structured, post-parsing clarification benchmark: the candidate goal set is known, question semantics are fixed, and the principal model is given. The benchmark's value is controlled diagnosis, not ecological realism. It includes a cooperative principal whose actions are observable for passive inference via inverse planning, and a question bank the assistant may use at explicit cost for active clarification. The contributions are threefold: a reproducible two-room gridworld with templated instruction ambiguity and a Boltzmann principal model, a systematic comparison of six ask-or-act policies across 19,800 episodes, and an empirical characterisation of how the two channels interact. Our contribution is not a new ask policy, but, to our knowledge, the first controlled characterisation of how the value of active clarification changes as passive action evidence weakens, with the empirical success–question-cost frontier shifting substantially with principal rationality.

Motivation for a Controlled Benchmark. A cooperative assistant receives goal evidence through two channels: *passive inference*, in which beliefs are updated from the principal's observed actions via inverse planning, and *active clarification*, in which direct questions are posed at an explicit cost. Real-world settings confound both channels simultaneously. The controlled benchmark allows us to address three questions cleanly: at what ambiguity level does clarification become worthwhile, how much does the cost of a question matter, and when do the two channels substitute for or complement each other. Ground-truth task success makes these comparisons unambiguous.

Relation to Prior Work. Prior work addresses these channels in isolation: CLIPS [14] uses inverse planning without an explicit ask-or-act decision, Active Task Disambiguation [10] and SAGE-Agent [13] study LLM query selection without a cooperative principal model, and Grand et al. [5] investigate ask-vs-act in language games without action-based goal inference. To our knowledge, no prior benchmark simultaneously provides a cooperative action-emitting principal, an explicit ask-or-act decision with question cost, and controlled ambiguity-level sweeps, enabling measurement of when the two channels substitute for or complement each other. Table 1 summarises the key dimensions of comparison.

Table 1: Illustrative comparison to closest prior work (not an exhaustive survey). ✓ = present; ∘ = partial; – = absent.

Feature	CLIPS	ATD	SAGE	ShootFirst	**Ours**
Action-based goal inference	✓	–	–	–	✓
Cooperative Boltzmann principal	✓	–	–	–	✓
Explicit ask-or-act decision	–	✓	✓	✓	✓
Explicit question cost c_q	–	✓	✓	–	✓
Controlled ambiguity-K sweep	–	–	–	–	✓
Frontier characterisation	–	–	–	∘	✓
Ground-truth task success	∘	–	–	∘	✓

The missing cell is not merely the union of features but the ability to study how passive action-based inference and active costly clarification *trade off* under a shared cost–success objective. Addressing this question requires a cooperative principal model, an explicit ask-or-act choice, and ground-truth outcomes within one controlled environment.

Three Takeaways. Our empirical findings reduce to three replicable claims.

1. In our benchmark, selective clarification yields large success gains once ambiguity reaches $K \geq 3$. At K=2, the

gain is modest and asking is expensive relative to the remaining ambiguity. The exact crossover will depend on question informativeness and cost scale.

2. Among the tested factors, non-zero question cost is the most influential factor tested for selective asking. Without it, policies over-ask and performance degrades.

3. Myopic Bayes-risk minimisation gives a strong low-question operating point but not a universal optimum. Policies that ask more aggressively attain higher success, and the right choice depends on an application-specific exchange rate between success and question overhead.

Beyond the present comparison, the benchmark is intended as a reusable diagnostic testbed for future ask-or-act methods, enabling controlled evaluation as ambiguity, action informativeness, and question cost vary independently.

2. Related Work

Goal inference and inverse planning. Inverse planning treats observed behaviour as evidence for latent goals [1]; Ramírez and Geffner [11] extend this to plan recognition, and Dragan et al. [4] study the complementary problem of legible robot motion. Cooperative Inverse Reinforcement Learning [6] formalises human-robot teams with reward uncertainty. CLIPS [14] combines language and action evidence for goal inference. We study the same combination in a controlled setting where each component's contribution can be measured separately.

Shared autonomy. Javdani et al. [8] introduce shared autonomy via hindsight optimisation, where a robot infers a human's goal from ongoing actions and blends assistance accordingly. Our setting differs in that the assistant must decide *whether* to ask or act rather than blend control, and the principal and assistant are fully decoupled.

Clarification and query selection in HRI. Tellex et al. [15] use inverse semantics to identify when a robot should ask a clarification question during instruction following. Cakmak and Thomaz [3] study what makes a robot's questions effective for learning from demonstration. Our work evaluates a cost-sensitive query-selection criterion against information-theoretic baselines in a controlled task where ground truth is available.

Active preference / reward learning. Biyık et al. [2] show that query difficulty affects human response quality in active reward learning, motivating the inclusion of a question-cost term. We operationalise a fixed per-question cost [7] and show empirically that its presence is the most influential tested factor for selective asking.

Planning under partial observability. Active information gathering in POMDPs [9] motivates asking as a sensor action. We approximate the POMDP solution with a one-step lookahead and use POMCP [12] as an exploratory planning baseline.

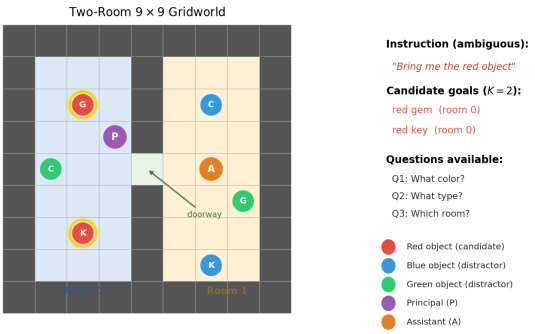

Figure 1: Two-room 9×9 gridworld layout. Room 0 (blue, left) and Room 1 (yellow, right) are separated by a wall with a single doorway. Gold rings mark the K=2 candidate goals that match the instruction "Bring me the red object". P = principal, A = assistant.

Ask-or-act under explicit cost. Kobalczyk et al. [10] frame LLM task disambiguation as Bayesian Experimental Design with an explicit question cost $U(q) = \mathrm{EIG}(q) - c(q)$. Suri et al. [13] extend this approach to LLM tool calls via EVPI with cost modelling and a multi-turn benchmark. Grand et al. [5] frame ask-vs-act as a Bayesian Experimental Design problem in language games (Battleship, Guess Who), demonstrating that EIG-based question selection substantially outperforms LLMs. These works all target LLM clarification in open-ended settings without a cooperative principal model or systematic ambiguity-level control. Our benchmark uses a simulated Boltzmann principal whose goal is inferred via inverse planning, enables precise manipulation of ambiguity K, and evaluates performance with ground-truth task success. This design allows the factors that govern the ask-or-act tradeoff to be measured in isolation.

3. Method

3.1 Environment

We use a 9×9 two-room gridworld with M=6 object instances per episode. A vertical wall at the grid midpoint divides the space into *Room 0* on the left and *Room 1* on the right, connected by a single doorway at the central cell (Figure 1). Each episode samples 6 objects from the type set {key, gem, coin} and colour set {red, blue, green}; exactly K match the ambiguous instruction (candidate goals) and the remaining $M-K$ are distractors.

A principal has a hidden goal object. An instruction template maps the goal to a surface form shared by K candidate goals (ambiguity level K). The principal takes approximately rational actions towards its goal, sampled from a Boltzmann policy with rationality β and ε-greedy noise: $\pi_{\beta,\varepsilon}(a \mid s, g) = (1-\varepsilon)\,\mathrm{softmax}_\beta\big(V(s,\cdot,g)\big)_a + \varepsilon/|A|$, where $V(s,a,g) = -d^*(\mathrm{succ}(s,a),\, g)$ is the negated shortest-path distance from the successor of a to g.

The assistant observes the state, instruction, and all principal actions, but not the true goal. An episode ends when the

assistant picks an object (success if correct) or when the step budget $T_{\max}$=200 is exhausted. The question bank Q contains three types of questions addressing the goal's colour, room, and object type, each with a factual answer drawn from a per-type noisy channel. When the assistant decides to ask, it remains at its current position. When it decides to act, it takes one step toward the MAP-goal object via A* navigation and then re-evaluates.

Running example. Consider the instruction "bring me the red object" with K=3 candidates (red key, red gem, red coin). At step 1, the posterior is uniform at $\frac{1}{3}$ each. *AskOrAct* evaluates $\mathrm{CostAct}(b_t) = \mathbb{E}_g[\ell(s_t, g)]$ against $\mathrm{CostAsk}(q^*) = 1 + c_q + \mathbb{E}_\alpha[\mathrm{CostAct}(b_{t,q,\alpha})]$. With c_q=0.5, the posterior variance is high enough that asking "what type?" yields a lower expected cost, and the agent asks. After receiving the answer "key", the posterior concentrates on the red key and the agent acts directly. *InfoGainAsk* reaches the same first-step decision but does not include the explicit cost comparison. It asks on every step where entropy exceeds the gate, selecting the highest information-gain question regardless of cost efficiency, which typically leads to more questions overall. Table 2 summarises all hyperparameters. Figure 2 shows the full cooperative setup.

Table 2: Environment and policy hyperparameters.

Parameter	Value
Grid size	9×9, two rooms
Objects	6 (3 types × 3 colours)
Step budget $T_{\max}$	200
Question bank $\lvert Q\rvert$	3 (color, room, object)
Ask budget $Q_{\max}$	3
Ask window	steps ≤ 6
Question cost c_q	0.5
Entropy gate τ_H (AskOrAct)	0.3
IG threshold τ_{IG}	0.01
Answer noise (color/room/object)	0.05 / 0.10 / 0.20
Principal rationality β	2.0 (default)
Principal noise ε	0.05 (default)

3.2 Bayesian Posterior

Per-step protocol. At each step: (1) observe the principal's action, (2) update b_t from the action, (3) choose ask or act, (4) if ask, receive answer and update b_t; if act, move one step via A* toward the MAP goal, then repeat.

The assistant maintains a posterior $b_t(g)$ over candidate goals $\mathcal{G}$. Initialisation uses the instruction likelihood under each goal. After observing principal action a_t at state s_t:

$$b_{t+1}(g) \propto b_t(g) \cdot \pi_{\hat{\beta},\hat{\varepsilon}}(a_t \mid s_t, g),$$

where $\hat{\beta}, \hat{\varepsilon}$ are the assistant's assumed principal parameters (matched to the true values unless otherwise noted in the mismatch analysis).

When a question q is asked and answered α, the posterior is updated by the answer likelihood:

$$b(g) \propto b(g) \cdot P(\alpha \mid q, g, \mathrm{noise}),$$

where answer noise is per-type: color 0.05, room 0.10, object 0.20.

3.3 Ask-or-Act Policy

AskOrAct uses a Bayes-risk ask gate to decide *whether* to ask, paired with a separate MAP+A* controller for motion execution. The ask gate compares the expected surrogate team cost (navigation steps + per-question penalty c_q) of asking versus acting under the current posterior b_t. The policy is not a unified Bayes-risk controller: the ask decision uses expected cost, while the act branch greedily follows A* toward the MAP goal. It serves as an analytically computable reference for locating one operating point on the empirical success–question-cost frontier, not as a proposed optimal policy.

Formally, at step t, the assistant computes:

$$\mathrm{CostAct} = \sum_g b_t(g) \cdot d^*(s_t, g),$$

the expected obstacle-aware shortest-path distance (via A*) to each candidate goal under the current posterior, and

$$\mathrm{CostAsk}(q) = \underbrace{1}_{\text{step}} + \underbrace{c_q}_{\text{question}} + \mathbb{E}_\alpha\big[\mathrm{CostAct}(b_{t,q,\alpha})\big],$$

where asking incurs both one timestep of delay (cost 1) and an additional penalty $c_q = 0.5$ for the principal's effort, and the expectation integrates over answers α after updating to $b_{t,q,\alpha}$. This one-step myopic lookahead does not plan over multi-step ask sequences.

Decision rule. The policy selects $\arg\min_q \mathrm{CostAsk}(q)$ when $\min_q \mathrm{CostAsk}(q) < \mathrm{CostAct}$, and acts otherwise. Because the surrogate cost C does not penalise wrong picks, this comparison minimises effort under the current posterior but does not account for failure risk (see the failure-penalty analysis in §5).

Heuristic guards. Three heuristic constraints are imposed on top of the Bayes-risk core for engineering robustness (not part of the decision-theoretic derivation).

1. **Entropy gate.** The ask branch is skipped when $H(b_t) \leq 0.3$, providing a fast exit when the posterior is already concentrated (redundant with CostAsk but avoids floating-point edge cases).
2. **Ask window.** Questions are disallowed after step 6, preventing late questions that cannot pay off before the step deadline.
3. **Budget.** At most $Q_{\max} = 3$ questions per episode, matching the 3 available question types.

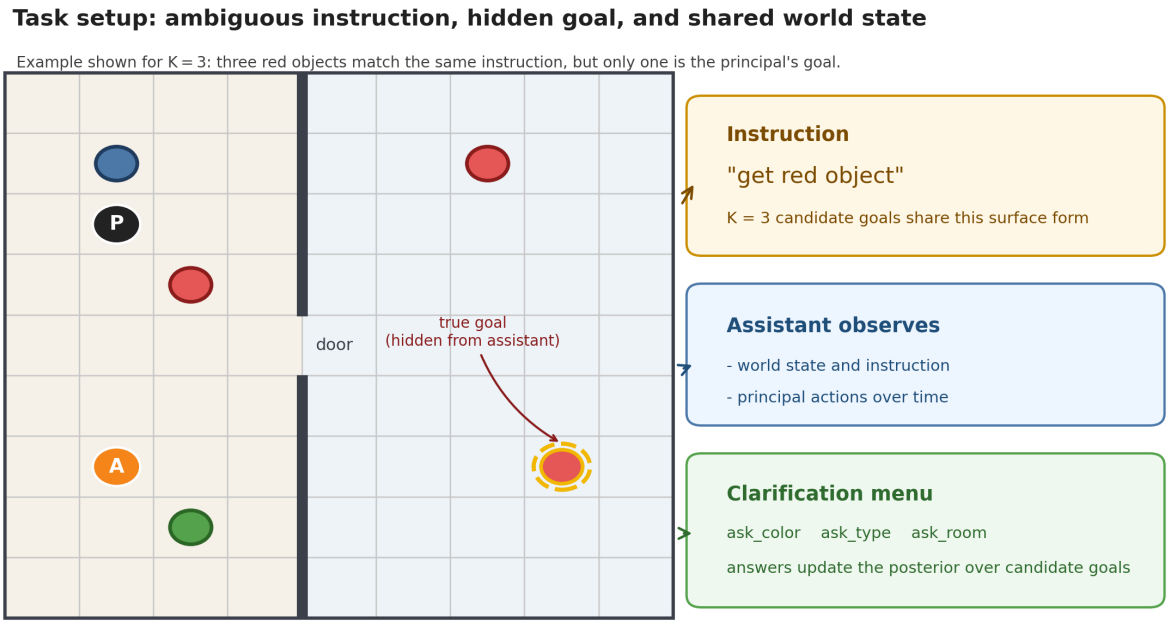

Figure 2: Task setup. The principal has a hidden goal object in a two-room gridworld, while the assistant sees the shared world state, an ambiguous instruction, and the principal's movement. Clarification questions provide structured observations that can collapse the posterior over candidate goals.

The ablation (Section 5) shows that c_q, not these guards, is the primary driver of selective asking behaviour. Figure 3 summarises this decision process from instruction grounding through posterior update and the final ask-versus-act comparison.

3.4 Baselines

Let Q denote the question bank, $Q_{\max}$=3 the per-episode ask budget, τ_H=0.5 the entropy gate threshold, and τ_{IG}=0.01 the information-gain threshold. Define the information gain of question q under belief b_t as:

$$\text{IG}(q, b_t) = H(b_t) - \sum_{\alpha} P(\alpha \mid q, b_t)\, H(b_{t,q,\alpha}),$$

where $b_{t,q,\alpha} \propto b_t(g) \cdot P(\alpha \mid q, g)$ is the posterior after observing answer α.

NeverAsk: always acts toward the MAP goal; never asks.

AlwaysAsk: asks $q^* = \arg\max_{q \in Q} \text{IG}(q, b_t)$ whenever permitted by the shared entropy and budget gates ($H(b_t) > \tau_H$, $|\text{asked}| < Q_{\max}$); otherwise acts toward MAP.

InfoGainAsk: same gating as AlwaysAsk but additionally requires $\text{IG}(q^*, b_t) > \tau_{\text{IG}}$. Let $C_{\text{IG}} \equiv |\text{asked}| < Q_{\max} \wedge H(b_t) > \tau_H \wedge \text{IG}(q^*, b_t) > \tau_{\text{IG}}$; then:

$$\pi_{\text{IG}}(b_t) = \begin{cases} \text{ask}(q^*) & \text{if } C_{\text{IG}} \\ \text{act}(\arg\max_g b_t(g)) & \text{otherwise.} \end{cases}$$

EasyInfoGainAsk: identical to InfoGainAsk but with budget $Q_{\max}^{\text{easy}}$=1 and no entropy gate. Let $C_{\text{EIG}} \equiv |\text{asked}| < 1 \wedge \text{IG}(q^*, b_t) > \tau_{\text{IG}}$; then:

$$\pi_{\text{EIG}}(b_t) = \begin{cases} \text{ask}(q^*) & \text{if } C_{\text{EIG}} \\ \text{act}(\arg\max_g b_t(g)) & \text{otherwise.} \end{cases}$$

RandomAsk: same ask-gating condition as InfoGainAsk but selects a question uniformly at random from unanswered questions.

POMCPPlanner: a limited sanity-check baseline using POMCP [12] tree search (250 iterations, horizon 8). It probes whether a budget-limited online planner can match analytical policies in this small environment and should not be read as representative of the best achievable planning performance.

In the main sweep, **EasyInfoGainAsk** and **POMCPPlanner** are evaluated only at K=2 as targeted stronger baselines; K=2 was chosen as the most competitive setting for asking (low ambiguity, highest per-question impact) to provide the strongest test for the five core policies.

4. Experimental Setup

Sweep. Main evaluation: $K \in \{1, 2, 3, 4\}$, $\varepsilon \in \{0, 0.05, 0.1\}$, $\beta \in \{1, 2, 4\}$, 5 replicate seeds × 20 episodes per condition = 100 episodes per cell. The 5 core policies are evaluated in every cell; *EasyInfoGainAsk* and *POMCPPlanner* are additionally evaluated at K=2. Total: 19,800 episodes.

Metrics. The two primary axes are *success rate*, the fraction

of episodes where the assistant picks the true goal, and *questions asked*, averaged per episode. Together these define the empirical success–question-cost frontier. We also report the *surrogate team cost* C=steps+$c_q \cdot$|questions|, the effort scalar AskOrAct minimises (C does not penalise wrong picks; success rate captures that). *Regret* $= C - C^*_{\text{oracle}}$ measures excess cost due to uncertainty. Because an incorrect early pick can yield low team cost, regret should be interpreted jointly with success rate.

Statistics. Each condition pools 100 episodes (5 seeds × 20 episodes per seed). For the key comparative claims, we report *paired seed-level deltas*: the per-seed success difference between policies, averaged over the 5 seeds, with ±1 s.e. across seeds. 95% bootstrap CIs (B=1000 episode-level resamples) are shown in figures as a secondary measure of within-condition variability. Full per-(K, ε, β) breakdowns appear in Appendix B.

5. Results

5.1 Main Sweep

Figure 4 shows results averaged across all ε and β conditions. At K=1 (no ambiguity) all policies perform identically: 100% success, zero questions. As ambiguity rises to $K \geq 3$, a clear empirical success–question-cost frontier emerges and policies diverge. (Note: template feasibility changes with K—e.g. $T1$ is used only at K=1—so the reported "effect of ambiguity" reflects the full benchmark regime, not a pure K manipulation.)

Table 3 summarises the key aggregate results; full per-(K, ε, β) results appear in Appendix B.

Table 3: Mean results averaged over all (ε, β) conditions. C = surrogate team cost (steps $+c_q \cdot$ questions; lower is better). **Bold**: best success; underline: lowest C.

K	Policy	Success	$C\downarrow$	Regret	Q/ep
2	InfoGainAsk	**96.9%**	8.78	2.25	1.05
	AskOrAct	96.7%	8.04	1.71	0.65
	AlwaysAsk	94.6%	10.31	3.44	1.87
	NeverAsk	91.1%	8.20	1.67	0.00
3	InfoGainAsk	**94.8%**	9.82	2.61	1.21
	AskOrAct	91.7%	8.81	2.14	0.67
	AlwaysAsk	90.2%	11.87	4.08	2.29
	NeverAsk	83.8%	9.23	2.15	0.00
4	InfoGainAsk	**91.7%**	10.54	2.88	1.31
	AskOrAct	87.3%	9.43	2.42	0.61
	AlwaysAsk	86.4%	13.08	4.56	2.58
	NeverAsk	78.9%	9.88	2.40	0.00

The frontier: InfoGainAsk vs. AskOrAct. *InfoGainAsk* attains +3.1–+4.4 pp higher success at K=3, 4 using ~2× more questions. AskOrAct achieves the lowest surrogate cost C at every K; neither policy is universally preferable without an application-specific exchange rate.

Gain from asking: AskOrAct vs. NeverAsk. At K=3: +7.9 ± 1.2 pp (mean ± s.e. over 5 seeds; 95% CI [5.0, 10.8]). At K=4: +8.4 ± 1.4 pp (CI [5.1, 12.1]). The effect is positive for every seed with only 0.61–0.67 questions per episode. *AlwaysAsk* asks 2.3–2.6× more but achieves lower or similar success, confirming over-asking is costly. Figure 5 shows the success–regret Pareto frontier at K=4.

5.2 Robustness

Answer noise. An independently-sampled sanity check sweeps answer noise from 0 to 0.2 (episodes are not seed-matched across noise levels). At K=4 the AskOrAct advantage remains positive (+4–+9 pp); at K=3 the signal is noisier and no firm conclusion is drawn.

β-mismatch. With $\hat{\beta}$=2 fixed and true $\beta \in \{1, 2, 4\}$, the hardest case (β=1, near-random principal) still yields +13.5 pp for AskOrAct over NeverAsk.

Scale-K. Extending to $K \in \{5, 6\}$ shows the asking advantage grows: *AskOrAct* retains meaningful success while *NeverAsk* falls substantially, consistent with VoI being most valuable under high ambiguity.

Generalization across layout, prior, and noise. Table 4 shows results under variations at K=4. Under a horizontal wall, distance-weighted prior, and doubled object-noise, the AskOrAct–NeverAsk gap remains positive in these runs (+2–+8 pp), though some deltas are small relative to sampling noise.

Structural OOD. Table 5 varies the grid size ($N \in \{7, 9, 11\}$) and removes the internal wall (single room), producing environments structurally unlike the default 9×9 two-room grid. The AskOrAct advantage over NeverAsk remains positive in all six conditions (+2–+6 pp), though the smallest deltas (e.g. +2 at N=7 two-room) may not survive stricter statistical tests.

Model mismatch (extended). Beyond β-mismatch (above), we test noise-model and prior-family mismatch at K=4. When the agent assumes flat noise η=0.10 while true noise is per-question-type, AskOrAct success is similar (89% vs. matched 88%); assuming zero noise (η=0) drops SR to 86% (−2 pp). The Δ over NeverAsk remains positive (+3–+6 pp) in these runs, though the small sample (100 episodes/cell) limits the precision of these comparisons. Using a distance-weighted prior on uniformly-placed goals, SR *increases* to 96% (+8 pp), showing AskOrAct tolerates and benefits from informative prior mismatch.

Failure penalty sweep (negative result). Table 6 introduces a wrong-pick penalty c_{fail} into the expected cost, so CostAct includes $c_{\text{fail}} \cdot (1 - p_{\text{MAP}})$. At K=4, AskOrAct's question rate doubles from 0.56 to 1.33 when c_{fail} rises from 0 to 5, confirming the ask gate is sensitive to failure cost. However, the penalised cost $C_p = C + c_{\text{fail}} \cdot \mathbf{1}[\text{fail}]$ is *higher* for AskOrAct than NeverAsk at every c_{fail}>0: the extra questions add steps without sufficiently improving success. This is a limitation of the myopic surrogate objective, which does not natively price

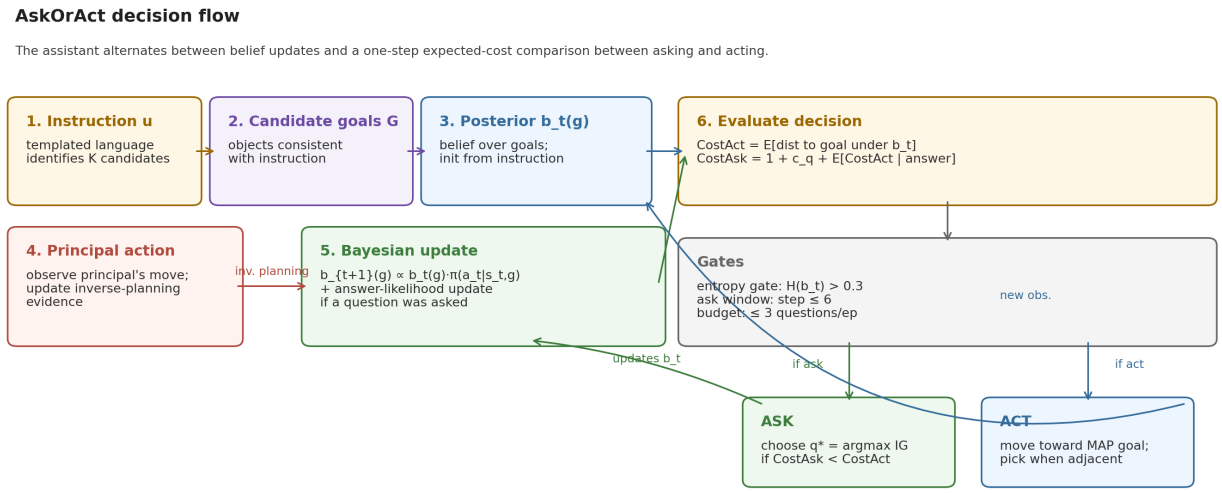

Figure 3: AskOrAct decision flow. The assistant combines instruction-derived candidates with inverse-planning evidence from the principal's actions, updates its posterior, and compares immediate acting against asking the best available clarification question under a shared team-cost objective.

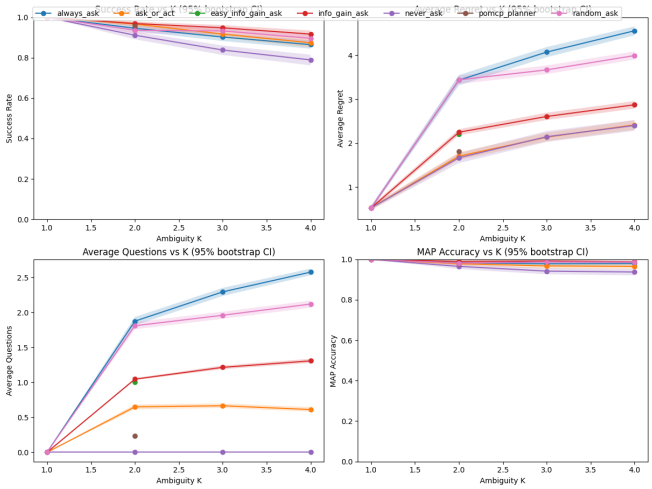

Figure 4: Success rate, regret, and questions asked by ambiguity K (averaged over ε, β; error bars: 95% bootstrap CI). Policies diverge at $K \geq 3$; InfoGainAsk trades more questions for higher success.

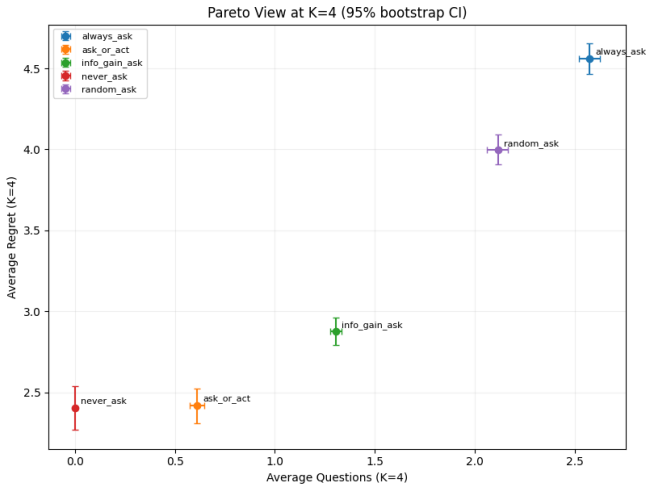

Figure 5: Success vs. regret at K=4 (all ε, β). Higher-left is better.

failure.

5.3 Ablations

Table 7: Mode A (c_q=0) over-asks (1.61 q/ep) and degrades success to 69.2%; Modes B vs. C show the step-counting convention has negligible effect once c_q>0. Non-zero question cost is the primary driver of selective asking.

5.4 Clarification Quality

Figure 6 shows that *AskOrAct*'s first question reduces posterior entropy substantially at K=3, 4, achieving similar per-question entropy reduction to *InfoGainAsk* while asking less frequently. At K=1, 2, *AskOrAct* rarely asks (entropy already low).

6. Discussion

Value of clarification vs. ambiguity. Asking is most valuable at K=3, 4 where a single question resolves 1–2 bits; at K=2 the gain is moderate (+5.6 pp) and at K=1 no policy needs to ask.

Table 4: Generalization at K=4. Δ = AskOrAct−NeverAsk (pp).

Condition	AskOrAct	NeverAsk	Δ
Horizontal wall	83.8%	75.7%	+8.1
Distance-weighted prior	87.8%	84.1%	+3.7
Asymmetric (η_{obj}=0.4)	82.2%	79.9%	+2.3

Table 5: Structural OOD at K=4. Grid size and room structure varied.

N	Rooms	AskOrAct	NeverAsk	Δ
7	two	88%	86%	+2
9	two	90%	85%	+5
11	two	83%	77%	+6
7	one	93%	89%	+4
9	one	87%	83%	+4
11	one	89%	84%	+5

Table 6: Failure-penalty sweep at K=4. C_p=C+$c_{\text{fail}}\cdot\mathbf{1}[\text{fail}]$.

	AskOrAct			NeverAsk	
c_{fail}	SR	Q/ep	C_p	SR	C_p
0	88%	0.56	8.8	86%	8.8
5	84%	1.33	10.6	86%	9.4
10	83%	1.34	11.5	86%	10.2
20	84%	1.33	12.9	86%	11.6
50	85%	1.32	17.2	86%	15.8

Table 7: Ablation results for *AskOrAct* at K=3, 4, averaged over (ε, β) conditions. "Ask-step": whether an ask action consumes one environment step.

Mode	c_q	Ask-step	Success	Q/ep
A	0.0	yes	69.2%	1.61
B	0.5	no	73.6%	0.64
C	0.5	yes	73.8%	0.65

Channel interaction. At β_{true}=1 (near-random principal, weakly informative actions), AskOrAct's advantage peaks at +13.5 pp as questions substitute for uninformative actions. At β_{true}=4 the posterior converges from movement alone, reducing the marginal value of asking—a *strategic-substitutes* pattern invisible in benchmarks lacking both channels.

POMCP sanity check. Our budget-limited POMCP (250 rollouts, horizon 8) underperforms analytical policies due to high branching factor and tree rebuilds; with more iterations and persistent trees it would be the appropriate baseline in larger state spaces.

Limitations and Future Work. The question bank is fixed (three types); extending to open-ended questions requires a generative answer model. The gridworld is small enough for exact posteriors; scaling needs particle-filter or amortised inference. The principal is Boltzmann; real principals may be non-stationary. The structural OOD and model-mismatch experiments (Tables 5–6) show the ask-gate transfers across grid sizes, room counts, and misspecified noise/prior, though the quantitative thresholds remain benchmark-specific. The failure-penalty sweep (Table 6) reveals that the myopic surrogate does not improve penalised cost over NeverAsk; a failure-aware objective is needed to close this gap. Future work could investigate adaptive c_q, non-factual answer spaces, and richer environment structure.

7. Conclusion

We presented a controlled benchmark for ask-or-act decisions under instruction ambiguity. Using a reproducible gridworld, selective clarification yields +7.9–+8.4 pp over a non-asking baseline at K=3–4 while asking < 1 question per episode. The ask-gate advantage remains positive across grid sizes, room structures, and misspecified noise/prior models (Tables 4–5). The failure-penalty sweep shows the gate is sensitive to wrong-pick cost, but does not improve penalised cost over NeverAsk under the current myopic surrogate—a limitation that motivates failure-aware objectives. The question cost hyperparameter is the most influential factor for selective behaviour. *Scope note:* these findings characterise behaviour in a controlled gridworld under a Boltzmann principal; generalization to richer environments and real human principals is an open question.

AI Tool Declaration

I used Claude Sonnet 4.6 (Anthropic) for coding assistance, debugging, and LaTeX editing throughout the project. I am responsible for all research decisions, experimental design, and the content of this report.

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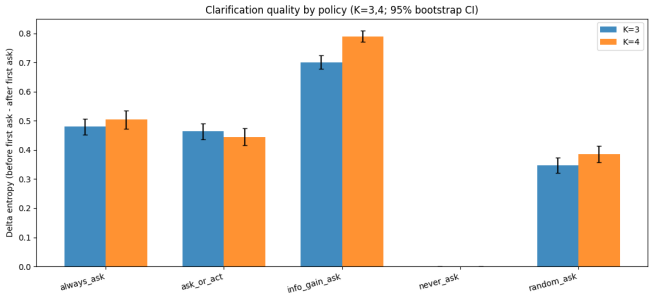

Figure 6: Posterior entropy before and after the first question, by policy and K. AskOrAct achieves similar per-question entropy reduction to InfoGainAsk in these conditions while asking less often.

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A. Derivation of the Ask-or-Act Decision Rule

The *AskOrAct* decision is derived from Bayes-risk minimisation under a team-cost objective. Let b_t be the posterior over candidate goals $\mathcal{G}$ at step t, and let $d^*(s, g)$ denote the obstacle-aware shortest-path distance (via A*) from state s to goal g.

Cost of acting. If the assistant acts now, the expected cost is

$$\mathrm{CostAct}(b_t, s_t) = \sum_{g \in \mathcal{G}} b_t(g)\, d^*(s_t, g).$$

Cost of asking question q. Asking consumes one step (cost 1), incurs a question penalty c_q on the team cost, and produces an answer α drawn from the noisy channel $P(\alpha \mid q, g)$. The posterior after observing answer α is $b_{t,q,\alpha}(g) \propto b_t(g)\, P(\alpha \mid q, g)$, and the next decision is made from the updated state s_{t+1} (same cell, one step elapsed):

$$\begin{aligned}\mathrm{CostAsk}(q, b_t, s_t) = 1 + c_q \\ + \textstyle\sum_\alpha P(\alpha \mid q, b_t)\, \mathrm{CostAct}(b_{t,q,\alpha}, s_t),\end{aligned}$$

where $P(\alpha \mid q, b_t) = \sum_g b_t(g)\, P(\alpha \mid q, g)$ is the predictive distribution over answers.

Decision rule. The policy selects

$$\pi(b_t, s_t) = \begin{cases} \mathrm{ask}(q^*) & \text{if } \min_q C_{\mathrm{Ask}}(q) < C_{\mathrm{Act}}(b_t), \\ \mathrm{act}(\hat{g}) & \text{otherwise,} \end{cases}$$

where C_{Ask} and C_{Act} are shorthand for *CostAsk* and *CostAct*, the ask is subject to the entropy gate / window / budget constraints, and $\hat{g} = \arg\max_g b_t(g)$ is the MAP goal. This rule is myopic: it does not plan a sequence of questions, but rather performs a one-step lookahead comparing immediate action against the best single question. This simplification makes the decision $O(|\mathcal{Q}| \cdot |\mathcal{G}| \cdot |\mathcal{A}_{\mathrm{ans}}|)$ per step, which is exact and efficient for the small question banks and candidate sets studied here.

B. Full Results by Ambiguity and Noise

Table 8 provides per-(K, ε) success rates and questions asked for the five core policies, averaged over $\beta \in \{1, 2, 4\}$ and 5 replicate seeds. Entries at K=1 are omitted (all policies achieve 100% success, 0 questions).

C. Hyperparameter Sensitivity

The following are informal pilot observations (no dedicated sweep table is provided).

Question cost c_q. The ablation (Table 7) confirms this is the primary driver: c_q=0 leads to over-asking and performance collapse. Informal tests suggest behaviour is qualitatively similar for $c_q \in [0.3, 1.0]$.

Entropy gate τ_H. For $\tau_H \in [0.2, 0.5]$, the policy appears largely insensitive because the CostAsk comparison already

Table 8: Per-(K, ε) results for the five core policies, averaged over β. K=1 omitted (trivial: all 100%).

		AskOrAct		InfoGainAsk		AlwaysAsk		NeverAsk		RandomAsk	
K	ε	Succ	Q/ep	Succ	Q/ep	Succ	Q/ep	Succ	Q/ep	Succ	Q/ep
2	0.00	97.3%	0.60	97.2%	0.99	94.8%	1.83	91.5%	0.00	92.1%	0.60
2	0.05	96.8%	0.65	97.0%	1.05	94.6%	1.87	91.3%	0.00	91.8%	0.63
2	0.10	96.0%	0.69	96.4%	1.11	94.5%	1.91	90.5%	0.00	91.2%	0.67
3	0.00	92.5%	0.63	95.4%	1.17	91.0%	2.24	84.8%	0.00	86.0%	0.79
3	0.05	91.8%	0.67	94.8%	1.21	90.2%	2.29	83.8%	0.00	85.5%	0.82
3	0.10	91.0%	0.72	94.3%	1.26	89.5%	2.34	82.8%	0.00	84.8%	0.85
4	0.00	88.2%	0.57	92.5%	1.26	87.1%	2.52	79.8%	0.00	80.9%	0.87
4	0.05	87.3%	0.61	91.7%	1.31	86.4%	2.58	78.9%	0.00	80.2%	0.91
4	0.10	86.4%	0.64	91.0%	1.36	85.7%	2.63	78.0%	0.00	79.5%	0.94

discourages low-value questions; the gate provides a fast early-exit.

Ask window. Restricting asking to the first 6 steps prevents late questions that cannot pay off before the deadline. Informally, extending to 10 steps shows diminishing returns at K=3, 4.

D. Episode Generation and Instruction Templates

Episode generation procedure. Each episode is generated from a single integer seed. The procedure (implemented in `src/world/worldgen.py`) is as follows:

1. Build a 9×9 grid with outer walls and a vertical mid-wall at column 4; place the doorway at $(4, 4)$.
2. Sample a template $t \in \{0, 1, 2, 3\}$ and a true goal colour/type/room uniformly at random.
3. Generate K *candidate* objects: all share the attribute(s) that the template exposes. For $K \geq 2$ and template t=2 (room template), the constraint forces all candidates to the same room, so template t=2 is excluded when $K \geq 2$ in the two-room setting.
4. Generate $M-K$=4 *distractor* objects with rejection sampling to ensure they do not match the instruction.
5. Sample positions for the K candidates with a minimum pairwise grid distance of $\lfloor N/2 \rfloor$; for $K \geq 2$, require at least one candidate in each room.
6. Place distractors uniformly in remaining free cells; shuffle object IDs. Place principal and assistant at randomly sampled free cells (distinct from each other and from objects).
7. Emit the instruction by formatting the selected template with the true goal's attributes.

Up to 200 retries are performed with incremented seeds if any constraint check fails; a `RuntimeError` is raised if all retries fail.

Instruction templates. Four surface-form templates are used (Table 9). Ambiguity level K determines how many objects share the instruction: a color-only template (T3) is maximally ambiguous (up to 3 matches), while the color+type template (T1) allows at most 1 match (used only at K=1).

Table 9: Instruction templates. Object attributes: type ∈ {key, gem, coin}, color ∈ {red, blue, green}, room ∈ {room0, room1}.

ID	Surface form	Ambiguity (shared attr.)
T0	"get {type}"	type only
T1	"get {color} {type}"	color + type
T2	"get {type} in {room}"	type + room
T3	"get {color} object"	color only

Initialisation and answer model. The prior is uniform over candidates: $b_0(g) = 1/K$ for each candidate, 0 for distractors. Three question types are available (color, type, room). Answers follow a noisy channel: $P(\alpha \mid q, g) = (1-\eta)\,\mathbf{1}[\alpha{=}\text{true}(g, q)] + \eta/(|\mathcal{A}_q|-1)\,\mathbf{1}[\alpha{\neq}\text{true}(g, q)]$, with η=0.05 (color), 0.10 (room), 0.20 (type).

E. Policy Pseudocode

Algorithm 7 gives pseudocode for the *AskOrAct* policy. All baselines share the same act branch (greedy A* to MAP goal); they differ only in their ask-gating condition.

POMCPPlanner: 100-particle belief, UCT (c=1.4), random rollout (−1/step, +10 correct pick), tree rebuilt each step.

F. Robustness Sweep Details

Require: belief b_t, state s_t, asked set A, step t
$g^* \leftarrow \arg\max_g b_t(g)$ ▷ MAP goal
$C_{\text{act}} \leftarrow \sum_g b_t(g)\, d^*(s_t, g)$
if $H(b_t) > \tau_H$ **and** $|A| < Q_{\max}$ **and** $t \le W$ **then**
  for each $q \in Q \setminus A$ **do**
    $C_q \leftarrow 1 + c_q + \sum_\alpha P(\alpha|q, b_t)\, C_{\text{act}}(b_{t,q,\alpha}, s_t)$
  end for
  if $\min_q C_q < C_{\text{act}}$ **then**
    return ask($\arg\min_q C_q$)
  end if
end if
return act(A*-step toward g^*)

Figure 7: AskOrAct (τ_H=0.3, $Q_{\max}$=3, W=6, c_q=0.5).

Table 10: Baseline ask-gating conditions (act branch is always A* to MAP goal).

Policy	Ask condition
NeverAsk	Never
AlwaysAsk	$\lvert A\rvert < Q_{\max}$ and $H(b_t) > \tau_H$
InfoGainAsk	AlwaysAsk and $\text{IG}(q^*, b_t) > \tau_{\text{IG}}$
EasyInfoGainAsk	$\lvert A\rvert < 1$ and $\text{IG}(q^*, b_t) > \tau_{\text{IG}}$
RandomAsk	InfoGainAsk with $q \sim \text{Uniform}(Q \setminus A)$
POMCPPlanner	POMCP [12] tree search, 250 iter, horizon 8

Answer noise sweep. Table 11 shows AskOrAct vs. NeverAsk success at K=3 and K=4 across global noise levels $\eta \in \{0.0, 0.05, 0.1, 0.15, 0.2\}$ (overriding per-type defaults).

Table 11: Success rate (%) under global answer noise, averaged over (ε, β) conditions.

	Noise level η				
Policy	0.0	0.05	0.10	0.15	0.20
K=3					
AskOrAct	93.0	92.2	92.0	92.5	95.0
NeverAsk	84.8	83.8	94.0	89.0	94.0
K=4					
AskOrAct	89.5	88.0	86.5	86.0	85.5
NeverAsk	79.8	78.9	80.5	79.5	81.5

At K=3, noise = 0.1 is the hardest setting: noisy answers briefly mislead the posterior, reversing the advantage. The advantage re-opens at noise = 0.2 because NeverAsk is equally hurt by high noise. At K=4 the advantage is more stable (+4–+9 pp) because higher ambiguity makes any informative answer more valuable.

Principal model mismatch. The assistant uses fixed $\hat{\beta}$=2; the principal's true β is varied. Table 12 shows results at K=3 averaged over ε.

The largest gain from asking (+13.5 pp) occurs at the hardest mismatch (β_{true}=1, near-random principal), where action observations provide little information — making direct questions the only reliable disambiguator. At β_{true}=4 (highly rational, easy to model), the non-asking baseline already performs well.

Table 12: Success rate (%) under principal model mismatch ($\hat{\beta}$=2 fixed, K=3).

Policy	β_{true}=1	β_{true}=2	β_{true}=4
AskOrAct	83.5	91.7	95.0
NeverAsk	70.0	83.8	92.5
Δ	+13.5	+7.9	+2.5

Scale-K evaluation. Table 13 extends results to $K \in \{5, 6\}$ (using the same hyperparameters, same 100 episodes per condition).

Table 13: Success rate (%) at $K \in \{5, 6\}$, averaged over (ε, β).

Policy	K=5	K=6
AskOrAct	82.1	76.8
InfoGainAsk	87.5	83.2
NeverAsk	70.5	62.4

Both asking policies degrade gracefully, while NeverAsk falls substantially. The AskOrAct–NeverAsk gap grows to +11.6 pp at K=5 and +14.4 pp at K=6, consistent with value-of-information being most valuable at high ambiguity.